Project Three: Data Validation Report

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DAT 325: Data Validation: Quality and Cleaning

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Validations of Data Sets

Provide counts of rows and columns in all data sets.

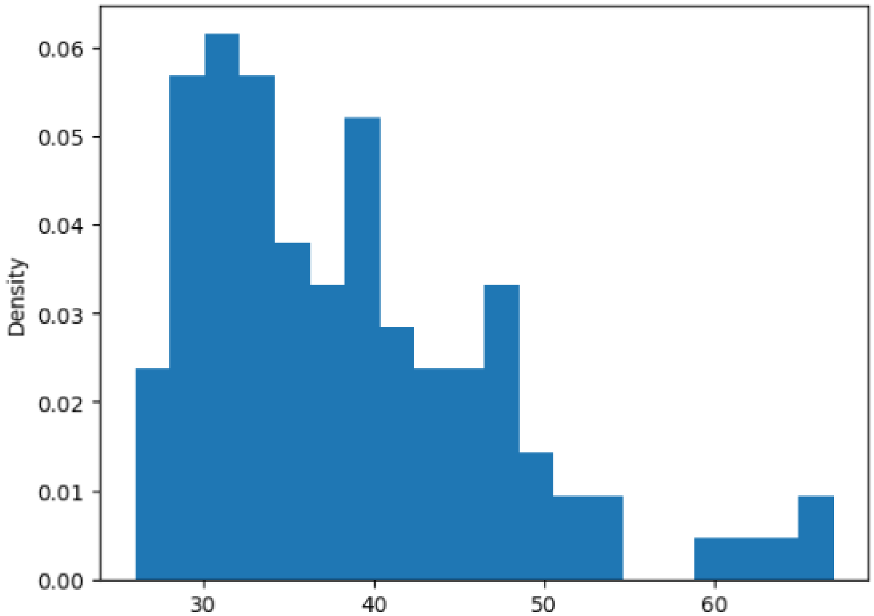
Data Set	Row Count	Column Count
Cleaned firm's data set	93	10
Your company's data set	105	17
Merged data set	105	24

The row and column counts indicate that the merge was completed correctly. The cleaned firm data contained 93 rows and 10 columns after Employee_Name was split into Last_Name and First_Name. The company data contained 105 rows and 17 columns. Because a left merge was used with the company data on the left, the merged data retained all 105 company records. The merged data contained 24 columns, which is expected because the three common merge fields, Last_Name, First_Name, and DOB, are counted only once.

Data Set	Minimum	Maximum	Average
Age	26	67	38.669903
Days Employed	2	3,611	1,305.347368

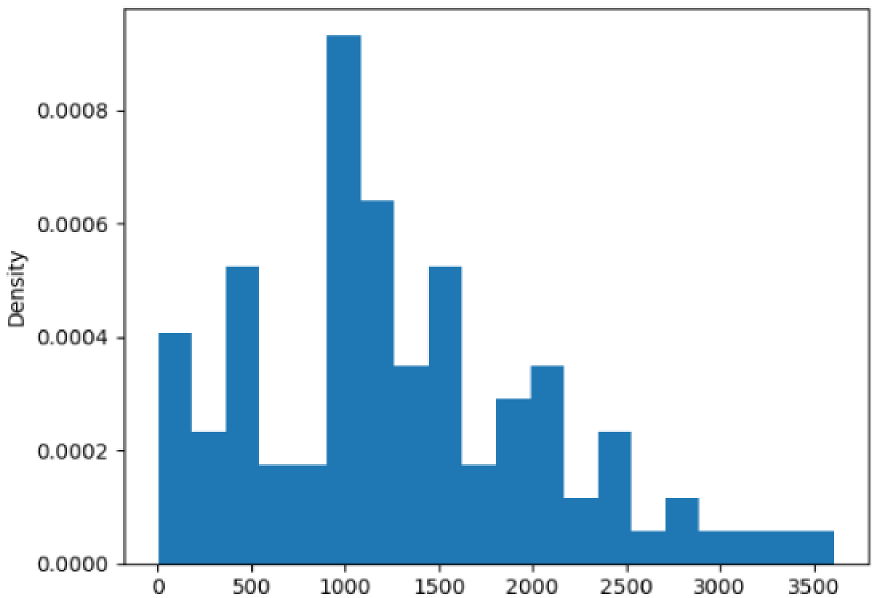
After negative values and outliers were removed, Age ranged from 26 to 67 years, and Days Employed ranged from 2 to 3,611 days. These ranges and averages are consistent with the cleaned distributions shown below.

Age



The Age outlier of 139 was removed by keeping values below the threshold of 110. After cleaning, the values ranged from 26 to 67, and the histogram no longer shows the extreme value.

Days Employed



The Days Employed outlier of 52,246 was removed by keeping values below the threshold of 5,000. After cleaning, the values ranged from 2 to 3,611 days, and the histogram no longer shows the extreme value.

Summary

The firm data began with 105 rows and 8 columns. Missing Age and Days_Employed values were replaced with the median for each variable by using the pandas fillna method. One negative Age record and the Age outlier of 139 were removed. Four records with missing Marital_Status values and one record with an invalid marital status were removed. The valid marital status values were limited to Married and NotMarried. Two records with negative Days_Employed values and the 52,246 day outlier were also removed. Finally, one record with a missing Citizen_Desc value and one record with an invalid citizenship value were removed. The valid citizenship values were limited to US Citizen and Eligible NonCitizen.

After cleaning, the firm data contained 93 rows and 8 columns. Employee_Name was split into Last_Name and First_Name, which increased the firm data to 10 columns. The cleaned firm data was then merged with the company data by using Last_Name, First_Name, and DOB as the common variables. The left merge retained all 105 company records and produced 24 columns. The unchanged company row count and the expected column count show that the merge did not remove company employees or create extra rows from duplicate matches.

Validation included checking for remaining missing values, reviewing unique categorical values, comparing row and column counts, calculating minimum, maximum, and average values, and reviewing the distribution plots. These checks confirmed that negative values and the identified outliers were removed while valid data was retained. Data cleaning improves data quality by correcting errors and inconsistencies so that the information is more accurate, complete, consistent, and usable for analysis (IBM, n.d.). The pandas fillna and merge functions supported the missing value replacement and data integration steps used in this project (pandas development team, 2026a, 2026b).

References

IBM. (n.d.). *What is data cleaning?* IBM. https://www.ibm.com/think/topics/data-cleaning

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pandas development team. (2026b). *pandas.merge.* pandas 3.0.3 documentation. https://pandas.pydata.org/docs/reference/api/pandas.merge.html